ANGEL ANDRÉS HERNÁNDEZ SERNA

Senior Industrial Engineer — Continuous Improvement — Automotive Manufacturing
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LinkedIn Profile — Professional Portfolio

Professional Summary

Senior Industrial Engineer with a proven track record in **automotive manufacturing** (JATCO México), complemented by additional experience in food manufacturing and fleet operations that broadened my expertise in process standardization, KPI management, and team leadership. Promoted directly from Jr to Senior Industrial Engineer within JATCO – skipping the Medium level – based on measurable performance impact, including a direct labor cost model presented to Nissan Japan leadership.

Track record includes:

- 98+ headcount positions optimized across production lines through time studies and process redesign
- +24% OEE increase
- -93% scrap reduction
- -25% corrective maintenance reduction
- 1,700+ monthly man-hours eliminated through automation

Specialized in Lean Manufacturing, Continuous Improvement, Workforce Planning, Industrial Automation, KPI Development and Process Optimization using VBA and Python, Plant Layout Design, and Ergonomic Assessments.

Professional Experience

ABC Autolineas — Saltillo, Coahuila, Mexico **Mar 2026 – Present**

General Supervisor – Preventive & Corrective Maintenance, Continuous Improvement

- Lead a cross-functional team of 14 (12 mechanics, 1 analyst, 1 preventive maintenance coordinator) across 3 repair facilities (Arteaga, Nuevo Laredo, Reynosa), overseeing a fleet of 251 units nationwide.
- Designed and built 3 KPI dashboards from the ground up (Work Orders, Preventive Maintenance, Corrective Maintenance), defining all underlying metrics and establishing real-time visibility where none previously existed.
- Reduced corrective maintenance interventions by 25% by implementing standardized preventive maintenance programs across the fleet.
- Achieved 92% unit-release compliance and 81% fleet availability across 251 units through standardized scheduling and 5S/visual management.

Flor de Calvillo — Calvillo, Aguascalientes, Mexico **Nov 2025 – Mar 2026**

Senior Industrial Engineer (Fixed-term Project Contract)

- Reduced labor requirements by 15 operators per shift through layout redesign and process optimization.
- Increased packaging productivity by supporting the implementation of an automated packaging and sealing process.
- Improved process stability by documenting and standardizing manufacturing operations using Lean methodologies.

JATCO México — Aguascalientes, Mexico **Jul 2022 – Nov 2025**

Senior Industrial Engineer *Mar 2025 – Nov 2025*

- Promoted directly from Jr to Senior Industrial Engineer, skipping the Medium level, after developing a direct labor cost segregation model by vehicle platform – presented to Nissan Japan leadership – which increased workforce efficiency by 60% monthly and enabled profitability on previously unprofitable projects.
- Reduced headcount by 44 operators across two cross-functional projects combining demand-based adjustments and continuous improvement initiatives, using man-machine analysis and time studies.
- Reduced casting scrap by 93% using root cause analysis and process control improvements.

Jr Industrial Engineer *Dec 2022 – Mar 2025*

- Led 5 headcount optimization projects across assembly, gear machining, heat treatment and pulley manufacturing areas, reducing labor requirements by 54 operators through time studies, motion analysis, ergonomic assessments, and line balancing.

- Designed and developed a dynamic workforce planning model in VBA that calculates headcount requirements based on production volume, OEE, and cycle time – eliminating 1,500+ monthly man-hours of manual calculation and standardizing capacity planning across production lines.
- Developed an automated Direct Labor Cost forecasting tool in VBA, reducing manual preparation time by 750 monthly man-hours.
- Led headcount and capacity planning for two new production line installations (machining, casting, material handling and warehouse areas), defining base times, standard times, plant layout distribution, and direct labor cost structure.

Quality Engineering Intern *Jul 2022 – Dec 2022*

- Reduced calibration tracking errors by implementing a VBA database system for calibration assets.
- Improved compliance visibility through automated reporting and calibration alerts.

Education

Full Stack Development in Python Diploma **2024 – Present**

EBAC Latam

Bachelor's Degree in Industrial Engineering **2021 – 2023**

Technological University of Aguascalientes

Postgraduate Degree in Automation and Quality **2020 – 2021**

Université de Lille, France

Technical University Degree in Industrial Processes – Manufacturing Area **2018 – 2020**

Technological University of Aguascalientes

Certifications

- Lean Six Sigma Yellow Belt
- Lean Productive Staff (JATCO México)
- Automotive Core Tools
- Certificate in Visual Basic for Applications (VBA) – Université de Lille, France (with Distinction)
- Leadership Development Program (JATCO México): Empathetic Leadership & Communication, Talent Management & Coaching, Emotional Intelligence & Conflict Resolution
- Full Stack Development in Python (In Progress)

Technical Skills

Manufacturing & Industrial Engineering

Lean Manufacturing, Continuous Improvement, Industrial Engineering, OEE, Kaizen, Time Studies, Motion Studies, Line Balancing, Capacity Analysis, Headcount Planning, Workforce Planning, Standard Work, Process Optimization, Cost Reduction, Direct Labor Cost, Team Leadership, Ergonomics, Plant Layout Design (LLD), Workstation Design.

Quality & Problem Solving

Root Cause Analysis, 5 Why, Ishikawa Diagram, PFMEA, CAPA, Process Control, Scrap Reduction, Corrective Actions.

Maintenance & Operations

Preventive Maintenance, Maintenance Planning, Fleet Maintenance, Operational Excellence, KPI Development, Maintenance Scheduling, Equipment Availability.

Digitalization & Automation

Advanced Excel, VBA Automation, Dashboard Development, KPI Automation, Python, Data Analysis, Reporting Automation.

Languages

- Spanish – Native
- English – B2 (Upper Intermediate)
- French – B2
- German – A1